UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta preliminare del 19/07/2018*

PROBLEM

It is given a random signal of interest (SOI) $S[n]$ which is modeled as an autoregressive (AR) process of order 1 with one-lag correlation coefficient ρ and power σ_S^2. This signal is observed in the presence of additive white Gaussian noise (AWGN) $W[n]$ with power σ_W^2. Hence, the observed signal is $X[n]=S[n]+W[n]$. $S[n]$ and $W[n]$ are mutually independent.

We want to design the optimal FIR Wiener filter of order P=2 to estimate $S[n]$ by processing the two observed samples $X[n\text{-}1]$ and $X[n\text{-}2]$.

(1) Derive the autocorrelation function (ACF) of the observed signal $X[n]$.

(2) Derive the impulse response $h[n]$ of the optimal FIR Wiener filter of order 2 and express it as a function of ρ and the signal to noise ratio (SNR) $\gamma \triangleq \sigma_S^2 / \sigma_W^2$. Find the expression of $h[n]$ when $\rho \to 1$ and explain the result.

(3) Derive the expression of the mean square error (MSE) of the optimal estimator $\hat{S}[n]$.

(4) Calculate numerically $h[n]$, the frequency response $H\left(e^{j2\pi f}\right)$, and the MSE when $\rho = 0.9$, $\sigma_S^2 = 2$, and $\sigma_W^2 = 1$. Plot the behavior of $h[n]$ and $\left|H\left(e^{j2\pi f}\right)\right|$, and establish what kind of filter it is (low-pass, high-pass, or band-pass).

(5) The value of the MSE would be greater or lower if $\rho = 0.5$? Provide a motivation to the answer.

Soluzione:

(1) La ACF del processo $S[n]$ è data da $R_S[m]=\sigma_S^2\rho^{|m|}$, quella di $W[n]$ è data da $R_W[m]=\sigma_W^2\delta[m]$, $S[n]$ e $W[n]$ sono processi indipendenti, quindi anche incorrelati, per cui risulta:

$R_X[m]=R_S[m]+R_W[m]=\sigma_S^2\rho^{|m|}+\sigma_W^2\delta[m]$.

(2) Filtro ottimo di Wiener FIR(2) per la stima di $S[n]$ dati gli osservati $X[n\text{-}1]$ e $X[n\text{-}2]$:

$$\hat{S}[n]=\sum_{k=1}^{2}h[k]X[n-k]=h[1]X[n-1]+h[2]X[n-2]$$

Principio di ortogonalità:

$$E\{\varepsilon[n]X[n-m]\}=E\{(S[n]-h[1]X[n-1]-h[2]X[n-2])\,X[n-m]\}=0,\quad m=1,2$$

$$R_{SX}[-m]=h[1]R_X[m-1]+h[2]R_X[m-2],\quad m=1,2$$

$$R_{SX}[-m]=R_S[m]$$

$$\begin{bmatrix} R_X[0] & R_X[-1] \\ R_X[1] & R_X[0] \end{bmatrix}\begin{bmatrix} h[1] \\ h[2] \end{bmatrix}=\begin{bmatrix} R_S[1] \\ R_S[2] \end{bmatrix}$$

$$\mathbf{h}=\begin{bmatrix} h[1] \\ h[2] \end{bmatrix}=\begin{bmatrix} R_X[0] & R_X[-1] \\ R_X[1] & R_X[0] \end{bmatrix}^{-1}\begin{bmatrix} R_S[1] \\ R_S[2] \end{bmatrix}=\begin{bmatrix} \sigma_S^2+\sigma_W^2 & \sigma_S^2\rho \\ \sigma_S^2\rho & \sigma_S^2+\sigma_W^2 \end{bmatrix}^{-1}\begin{bmatrix} \sigma_S^2\rho \\ \sigma_S^2\rho^2 \end{bmatrix}$$

$$=\begin{bmatrix} 1+\frac{1}{\gamma} & \rho \\ \rho & 1+\frac{1}{\gamma} \end{bmatrix}^{-1}\begin{bmatrix} \rho \\ \rho^2 \end{bmatrix}=\frac{\rho}{\left(1+\frac{1}{\gamma}\right)^2-\rho^2}\begin{bmatrix} 1+\frac{1}{\gamma} & -\rho \\ -\rho & 1+\frac{1}{\gamma} \end{bmatrix}\begin{bmatrix} 1 \\ \rho \end{bmatrix}=\frac{\rho}{\left(1+\frac{1}{\gamma}\right)^2-\rho^2}\begin{bmatrix} 1+\frac{1}{\gamma}-\rho^2 \\ \rho\left(1+\frac{1}{\gamma}\right)-\rho \end{bmatrix}$$

$$\mathbf{h}=\begin{bmatrix} h[1] \\ h[2] \end{bmatrix}=\begin{bmatrix} \dfrac{\rho\left(1+\frac{1}{\gamma}-\rho^2\right)}{\left(1+\frac{1}{\gamma}\right)^2-\rho^2} \\[2ex] \dfrac{\rho^2\frac{1}{\gamma}}{\left(1+\frac{1}{\gamma}\right)^2-\rho^2} \end{bmatrix}=\begin{bmatrix} \dfrac{\gamma\rho\left(1+\gamma-\gamma\rho^2\right)}{(1+\gamma)^2-\gamma^2\rho^2} \\[2ex] \dfrac{\gamma\rho^2}{(1+\gamma)^2-\gamma^2\rho^2} \end{bmatrix}$$

$$h[n] = \frac{\gamma\rho\left(1+\gamma-\gamma\rho^2\right)}{(1+\gamma)^2-\gamma^2\rho^2}\delta[n-1] + \frac{\gamma\rho^2}{(1+\gamma)^2-\gamma^2\rho^2}\delta[n-2]$$

$$\hat{S}[n] = \frac{\gamma\rho\left(1+\gamma-\gamma\rho^2\right)}{(1+\gamma)^2-\gamma^2\rho^2}X[n-1] + \frac{\gamma\rho^2}{(1+\gamma)^2-\gamma^2\rho^2}X[n-2]$$

$$\lim_{\rho\to 1} h[n] = \frac{\gamma(1+\gamma-\gamma)}{(1+\gamma)^2-\gamma^2}\delta[n-1] + \frac{\gamma}{(1+\gamma)^2-\gamma^2}\delta[n-2] = \frac{\gamma}{2\gamma+1}\cdot\left(\delta[n-1]+\delta[n-2]\right)$$
$$= \frac{1}{1+\frac{1}{2\gamma}}\cdot\frac{\delta[n-1]+\delta[n-2]}{2}$$

$$\lim_{\rho\to 1}\hat{S}[n] = \frac{1}{1+\frac{1}{2\gamma}}\cdot\frac{X[n-1]+X[n-2]}{2} = \alpha\overline{X}$$

Quando $\rho \to 1$ il segnale utile tende ad essere completamente correlato e lo stimatore di $S[n]$ tende allo stimatore di un segnale aleatorio (Gaussiano) costante S osservato in presenza di rumore AWGN. Come è noto, tale stimatore è dato dalla media campionaria $\overline{X}$ scalata dal coefficiente $\alpha \triangleq \frac{1}{1+\frac{1}{N\gamma}}$, in questo caso calcolato per N=2.

(3) MSE filtro ottimo di Wiener FIR(2):

$$MSE = \sigma_\varepsilon^2 = E\left\{\varepsilon^2[n]\right\} = E\left\{\varepsilon[n]S[n]\right\} = E\left\{\left(S[n]-h[1]X[n-1]-h[2]X[n-2]\right)S[n]\right\}$$

$$= R_S[0]-h[1]R_{SX}[-1]+h[2]R_{SX}[-2] = R_S[0]-h[1]R_S[1]+h[2]R_S[2]$$

$$= \sigma_S^2\left(1-h[1]\rho-h[2]\rho^2\right) = \sigma_S^2\left(1-\frac{\gamma\rho^2\left(1+\gamma-\gamma\rho^2\right)}{(1+\gamma)^2-\gamma^2\rho^2}-\frac{\gamma\rho^4}{(1+\gamma)^2-\gamma^2\rho^2}\right)$$

(4) $\rho = 0.9$, $\sigma_S^2 = 2$ e $\sigma_W^2 = 1$:

$$\mathbf{h} = \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} 0.4313 \\ 0.2813 \end{bmatrix}$$

$$H\left(e^{j2\pi f}\right) = \sum_{n=1}^{2} h[n] \cdot e^{-j2\pi fn} = h[1]e^{-j2\pi f} + h[2]e^{-j4\pi f} = \left(0.4313 + 0.2813e^{-j2\pi f}\right)e^{-j2\pi f}$$

$$\begin{aligned} \left|H\left(e^{j2\pi f}\right)\right| &= \left|0.4313 + 0.2813e^{-j2\pi f}\right| = \sqrt{\left(0.4313 + 0.2813\cos(2\pi f)\right)^2 + \left(0.2813\sin(2\pi f)\right)^2} \\ &= \sqrt{(0.4313)^2 + (0.2813)^2 + 2 \cdot 0.4313 \cdot 0.2813\cos(2\pi f)} \\ &= \sqrt{0.2651 + 0.2424\cos(2\pi f)} \end{aligned}$$

$$MSE = \sigma_S^2\left(1 - h[1]\rho - h[2]\rho^2\right) = 0.7680$$

(5) $\rho = 0.5$, $\sigma_S^2 = 2$ e $\sigma_W^2 = 1$:

$$\mathbf{h} = \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} 0.3125 \\ 0.0625 \end{bmatrix}$$

$$H\left(e^{j2\pi f}\right) = \left(0.3125 + 0.0625e^{-j2\pi f}\right)e^{-j2\pi f}$$

$$\begin{aligned} \left|H\left(e^{j2\pi f}\right)\right| &= \left|0.3125 + 0.0625e^{-j2\pi f}\right| = \sqrt{\left(0.3125 + 0.0625\cos(2\pi f)\right)^2 + \left(0.0625\sin(2\pi f)\right)^2} \\ &= \sqrt{0.1016 + 0.0391\cos(2\pi f)} \end{aligned}$$

$$MSE = \sigma_S^2\left(1 - h[1]\rho - h[2]\rho^2\right) = 1.6563$$

Se il segnale è meno correlato, la stima del segnale al passo successivo peggiora e l'MSE aumenta. In entrambi i casi il filtro è un passa-basso, però nel primo caso ($\rho = 0.9$) il filtro è a banda più stretta, il che è ragionevole in quanto anche il processo $S[n]$ è a banda più stretta.